



# **Gem car tracking application**

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**BACHELOR OF ENGINEERING**

**IN COMPUTER ENGINEERING**

**MAE FAH LUANG UNIVERSITY**

**2023**

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**UNIVERSITY**

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**ALONGKOD PANFU**

A COMPUTER ENGINEERING PROJECT SUBMITTED TO  
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF  
THE REQUIREMENTS FOR THE DEGREE OF  
BACHELOR OF ENGINEERING  
IN COMPUTER ENGINEERING

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# **Gem car tracking application**

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THIS COMPUTER ENGINEERING PROJECT HAS BEEN

APPROVED

TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS

FOR THE DEGREE OF BACHELOR OF ENGINEERING

IN COMPUTER ENGINEERING 2023

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## **ACKNOWLEDGEMENT**

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| Asst. Prof. Suppakarn Chansareewittaya | Committee |

**Abstract**

This project presents about tracking gems in university where we have seen the problem of our university students not being able to keep track of time. Or know in advance when the gem car will come or the gem car will be full. We will therefore add features to the MFU life app or create a sign at the Gem car station telling the time that the car will arrive at each station and will follow the GPS tracker of each car to know the number of Gem car users. that in order to facilitate the life of university students and university personnel as well.

Keyword: GPS tracker, Gem car, application MFU life

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background and Rationale

1.1.1 GPS will notify you when the Gem bus arrives at the Gem bus station. Notifications will be sent to each person's LINE application and receive a Token to receive notifications.

### 1.2 Objective

1.2.1 in order to save travel time

1.2.2 Increase the convenience of university life.

### 1.3 Scope

1.3.1 General user Students and personnel in educational institutions

1. Show the GPS of the car.
2. There is a sign displayed at the bus station.

1.3.2 Administrator1)

1. List all admin functions

## **1.4 Methodology**

### 1.4.1 Proposal preparation and defense

1.4.1.1 Find the topic of the project and present it to the advisor and committee.

### 1.4.2 Literature review and requirement gathering

1.4.2.1 Find related research to compare with the topic the group has chosen and analyze their advantages and disadvantages.

### 1.4.3 Acquiring system requirements with potential users, i.e., university

1.4.3.1 Survey the area for installing the sign.

### 1.4.4 System evaluation (functionality and user assessment)

1.4.4.1 Finally taken from users to test the functionality and suggestions as well as comments user comments about this project.

### 1.4.5 System manual

1.4.5.1 Detailed instruction manual for user convenience

## 1.5 Plan

**Table 0.1** Working plan

| Plan/Month                                 | Pre-project |     |     |     |     | Project |     |     |     |     |
|--------------------------------------------|-------------|-----|-----|-----|-----|---------|-----|-----|-----|-----|
|                                            | Jan         | Feb | Mar | Apr | May | Aug     | Sep | Oct | Nov | Dec |
| Finding the topic and Proposal preparation |             |     |     |     |     |         |     |     |     |     |
| Research the information                   |             |     |     |     |     |         |     |     |     |     |
| System analysis                            |             |     |     |     |     |         |     |     |     |     |
| System design                              |             |     |     |     |     |         |     |     |     |     |
| Prepare for final presentation             |             |     |     |     |     |         |     |     |     |     |
| System development                         |             |     |     |     |     |         |     |     |     |     |
| Check and solve problem                    |             |     |     |     |     |         |     |     |     |     |
| Finish software / hardware                 |             |     |     |     |     |         |     |     |     |     |
| Documentation / Report                     |             |     |     |     |     |         |     |     |     |     |
| Final presentation                         |             |     |     |     |     |         |     |     |     |     |

## 1.6 Expected Result.

1.6.1 Reduce the problem of waiting for the gem car.

## 1.7 Place and Equipment

1.7.1 Equipment

1.7.1.1 Hardware (Identify also the minimum specification)

1.Arduino R3



## 2. GPS module 6M



## 3. ESP 32



### 1.7.1.2 Software (List all target software and version)

- Arduino IDE



### 1.7.2 Place (Locate the working place such as MFU and/or other places for getting requirements, testing, and installing the system)

1.7.2.1 The Gem bus station sign and the location of the university.

### 1.7.3 Budget

-Arduino r3 173 THB

-GPS module 6 m 130 THB

- ESP32 200 THB

= 503 THB

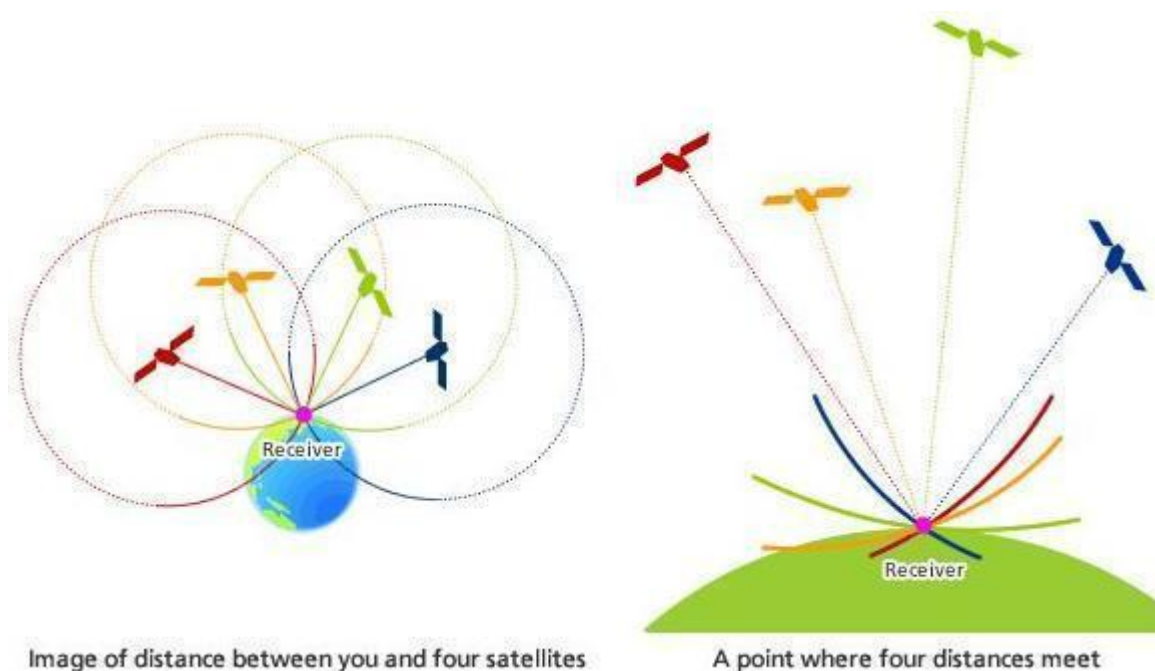
## CHAPTER 2

### LITERATURE REVIEW

#### 2.1 Related Theory

##### 2.1.1 Theory of Relativity with GPS Systems

The GPS system has a basic working principle. If we know how many meters away from each satellite we are. We will know where we are in the world. Therefore, each satellite in the sky will measure the distance from the GPS device to it by means of timing the radio wave signals emitted to the GPS device and processing the data to tell the position, latitude, longitude and height above water level. The sea of GPS devices, and even more in the sky, there are many satellites. Recording and matching the position of the device will be more accurate.



**Figure 2.1** The image shows the positioning of GPS devices from multiple satellites in the sky.

In addition, the positioning accuracy depends on the location of each satellite. That is, if the distance between active satellites is far apart, it will give more accurate values than those near each other. And the more satellites that can receive a signal, the more accurate it will be.

**Atmospheric variability** The atmosphere consists of electric charges, humidity, temperature and density that fluctuate over time. waves on impact with various objects Will cause refraction to weaken the signal. And the environment in the receiving area, such as being obscured by glass, water spray, leaves will affect the accuracy of the accuracy. Because if the signal from the satellite is refracted, it will cause the calculated value from the receiver to be distorted. And finally, it's the performance of the receiver, how sensitive it is to receive the signal and the processing speed.

### **2.1.2 The distance between the satellite and the receiver can be measured using**

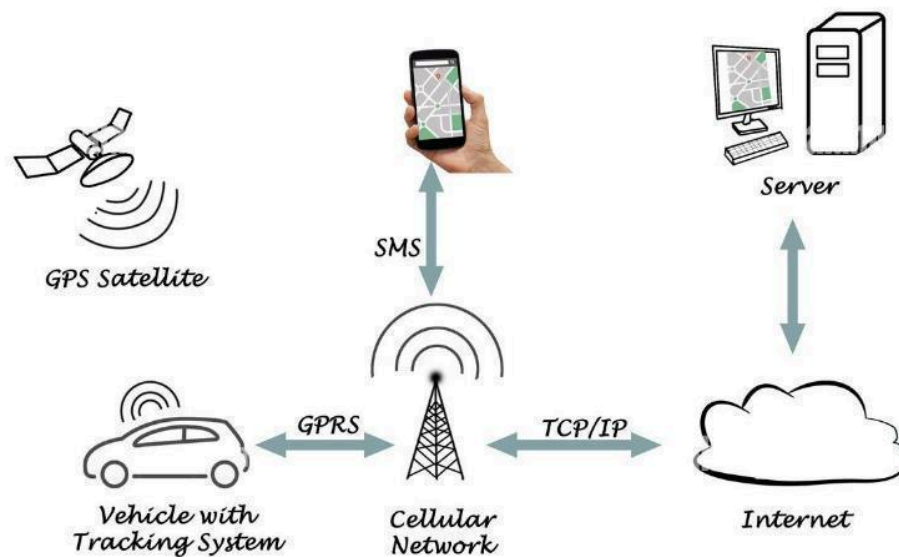
Distance = Speed \* Duration Measures the time that radio waves transmit from the satellite to the GPS receiver multiplied by the speed of the radio wave equals the distance that the receiver receives. away from the satellite The time is measured from a high-precision satellite clock with a resolution of nanoseconds. and always verified with ground stations

## 2.2 Related Work

### 2.2.1 car tracking

Improves the efficiency of car tracking which can accurately track the coordinates of the cars that are currently being used To assist in the management of the car to be used as efficiently as possible.

### *GPS Tracking Technology*



**Figure 2.2** GPS Tracking Technology

### 2.2.2 Help to track

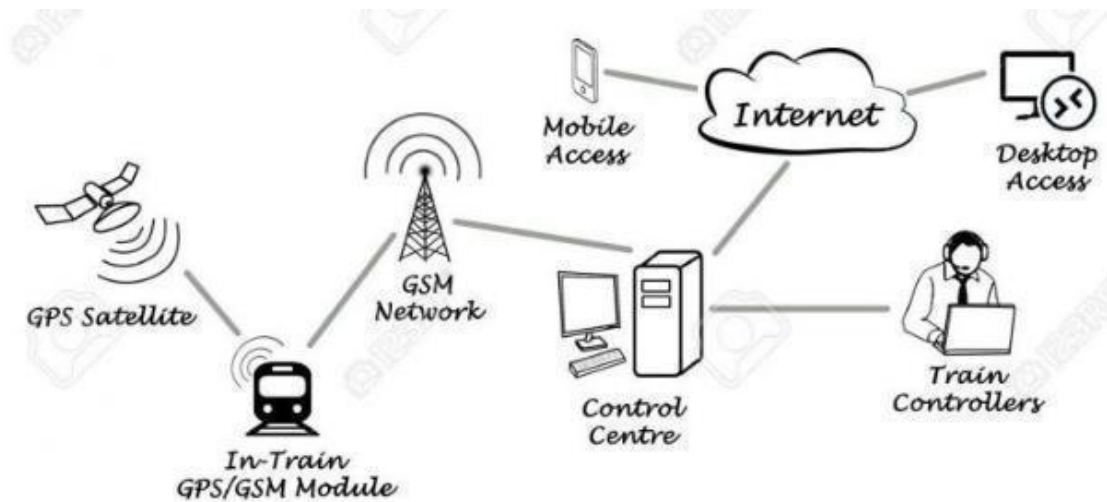
The work of the car in the transportation business. in a matter of time to ensure that all vehicles travel along the specified route no detour and travel safely to reach the destination on time.



## CHAPTER 3

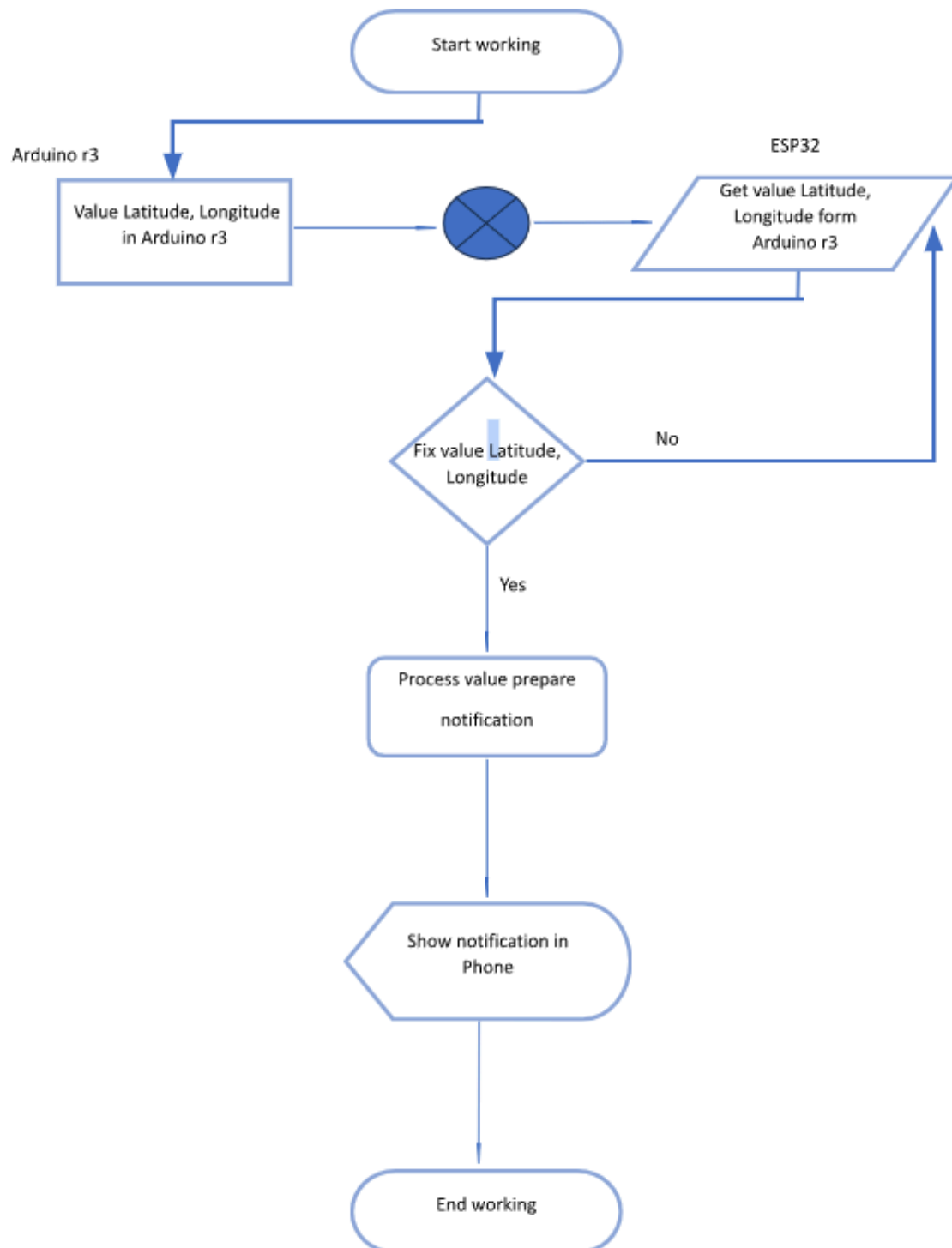
### ANALYSIS AND DESIGN

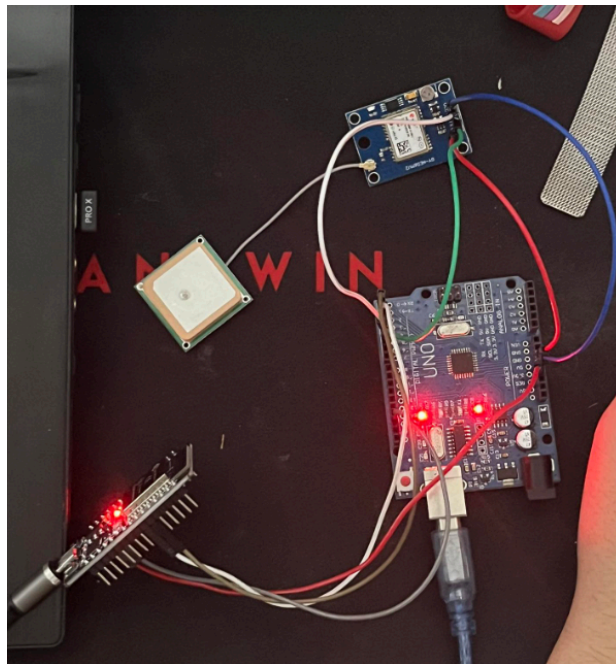
**By design** the system will be designed by receiving a signal from the GPS installed on each gem vehicle to enter into an application or a sign installed on each gem station and on campus.



**Figure 3.1** Real-time GPS Tracking

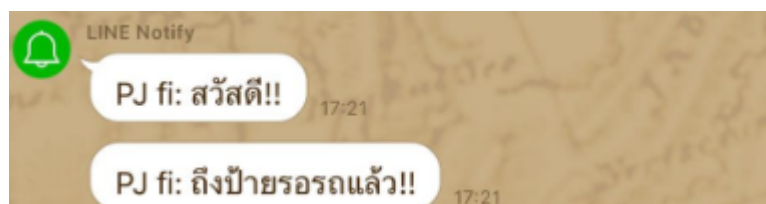
## Flowchart





**Figure 3.2** Connecting the Esp32 device to receive notifications

We connect to Wi-Fi so that the board receives Wi-Fi values and then write code to receive values of latitude and longitude in order to notify in Line.



**Figure 3.3** Notification via LINE

```

#define RXD2 18
#define TXD2 19
#include <TridentTD_LineNotify.h>

#define SSID      "FK"           // บรรทัดที่ 11 ไม่ใส่ ชื่อ Wifi ที่จะเชื่อมต่อ
#define PASSWORD  "0660198989"  // บรรทัดที่ 12 ไม่ ใส่ Wifi
#define LINE_TOKEN "m442TmRt0XhUksowym12HuntU70yEqE2s9zkWUv220" // บรรทัดที่ 13 ใส่ รหัส TOKEN ที่ได้จากทางบน

double targetLatitude = 20.049688;

void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  Serial.begin(115200);
  Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2);
  WiFi.begin(SSID, PASSWORD);
  Serial.printf("Wifi connecting to %s\n", SSID);
  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(400);
  }
  Serial.printf("\nWifi connected\nIP : ");
  Serial.println(WiFi.localIP());

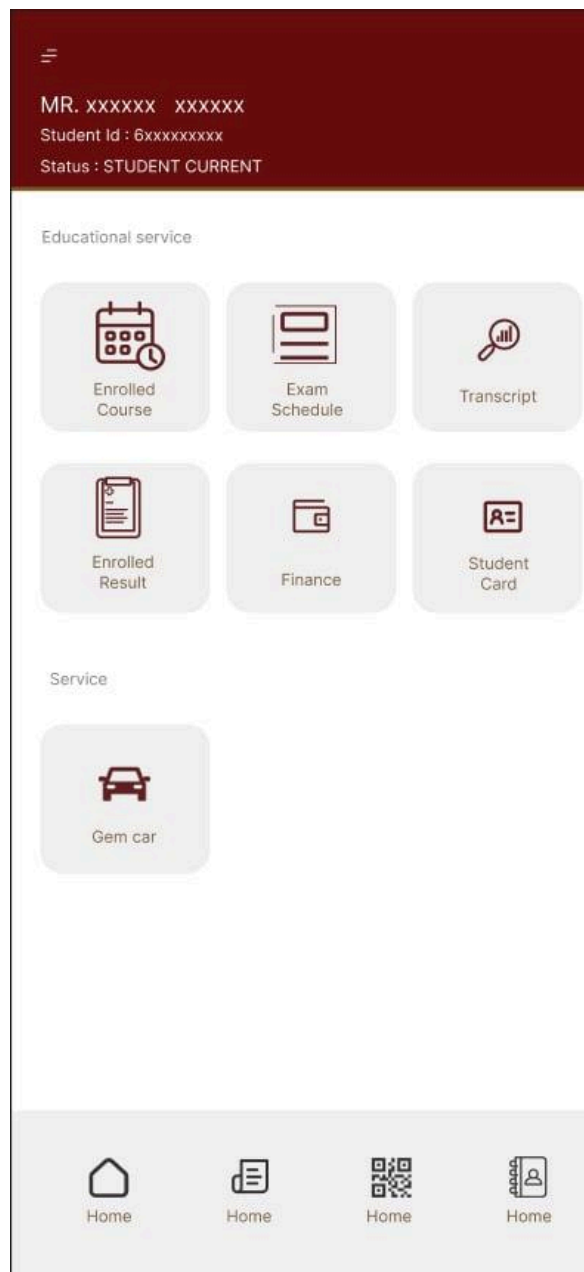
  // กำหนด Line Token
  LINE.setToken(LINE_TOKEN);
  // ส่งข้อความ
  LINE.notify ("สวัสดี!!");
}

void loop() {
  if (Serial2.readString()){
    String receivedData = Serial2.readStringUntil('\n');
    double newLatitude = receivedData.toDouble();
    //targetLatitude = Serial2.readString();
    if (newLatitude != targetLatitude) {
      targetLatitude = newLatitude;
      LINE.notify ("ถึงเป้าหมายรถแล้ว!!");
    }
  }
  return
  delay(100);
}

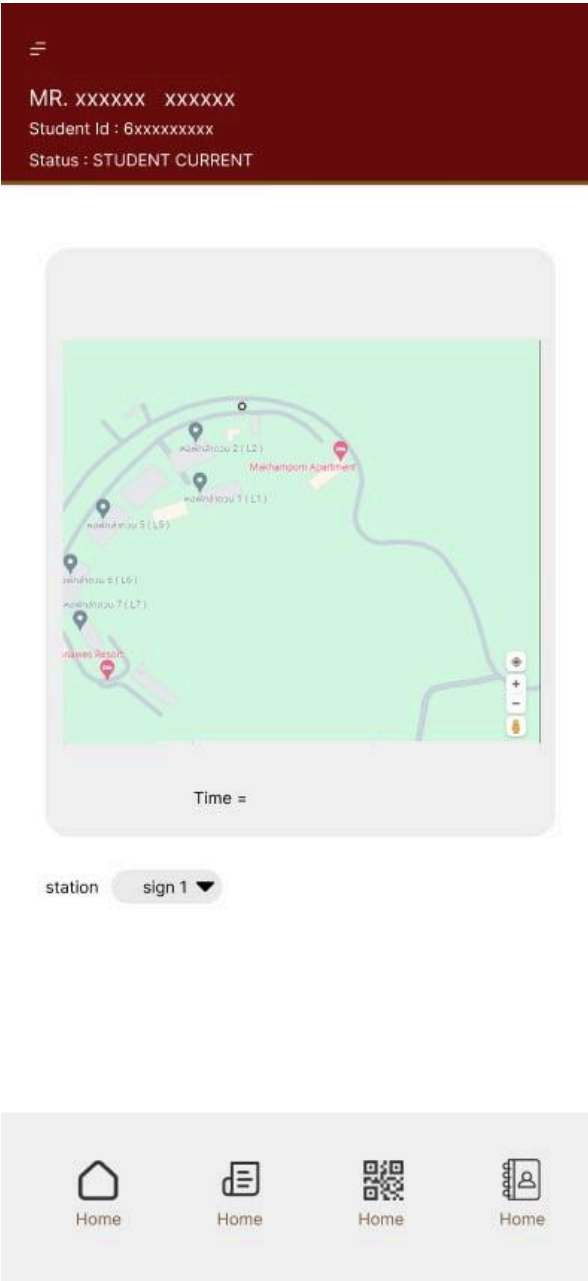
```

**Figure 3.4** Code for getting the values of the esp32 board

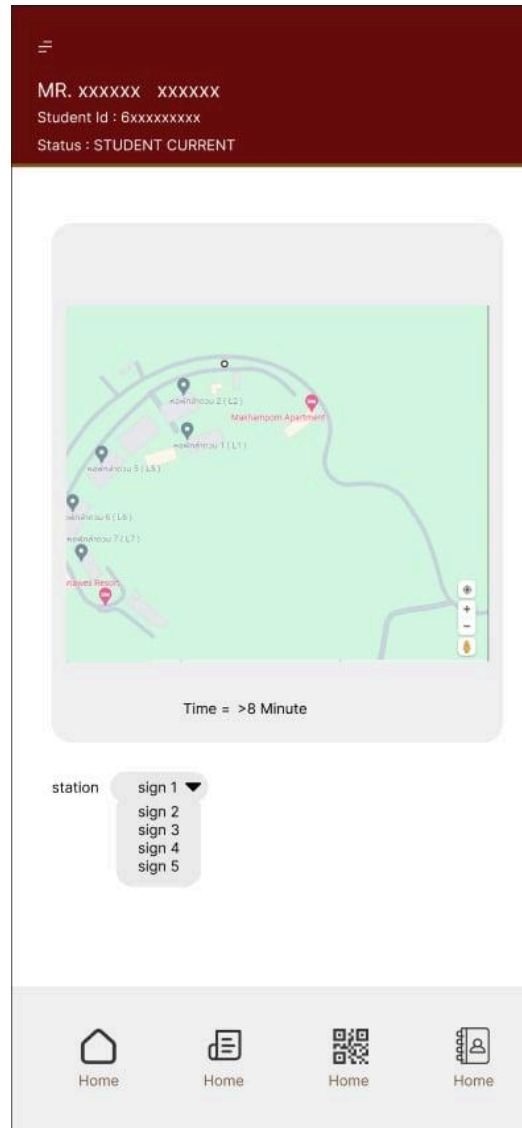
After we receive the value of Esp32, we set the value to the coordinates of the gem station. If the gem bus arrives at that station, it will send a notification to the phone connected to the line token. The system will notify you that the car has arrived for you.



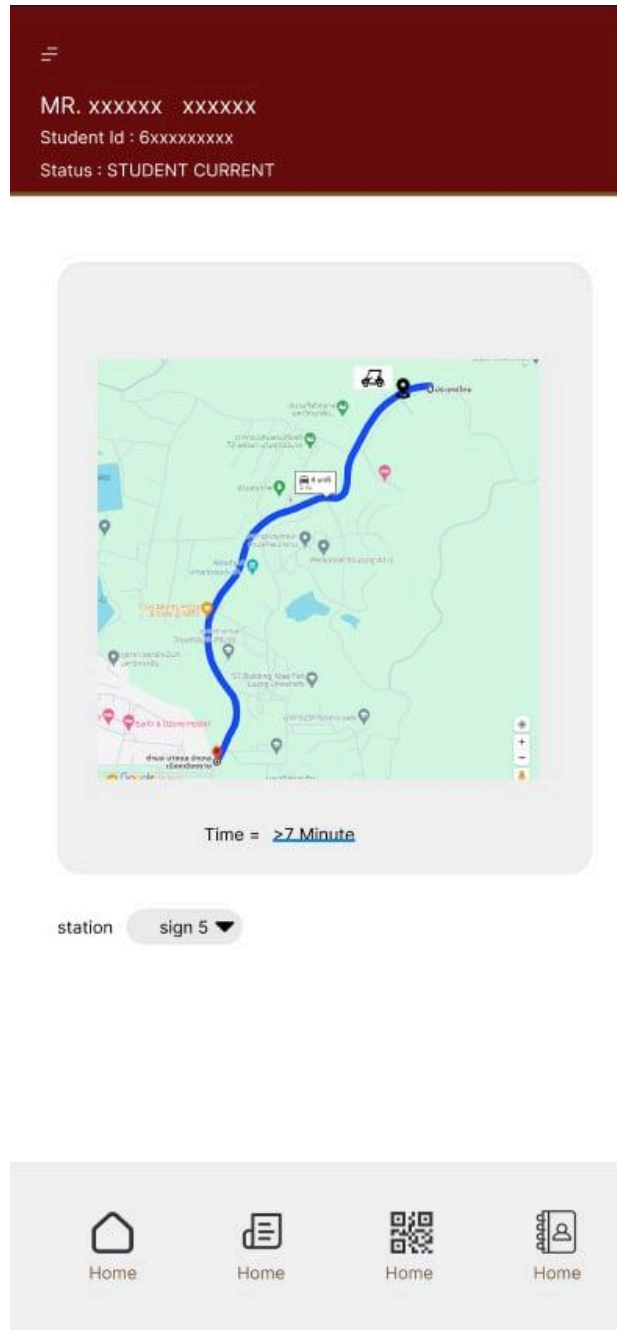
**Figure 3.5 display app**



**Figure 3.6 in icon gem car**

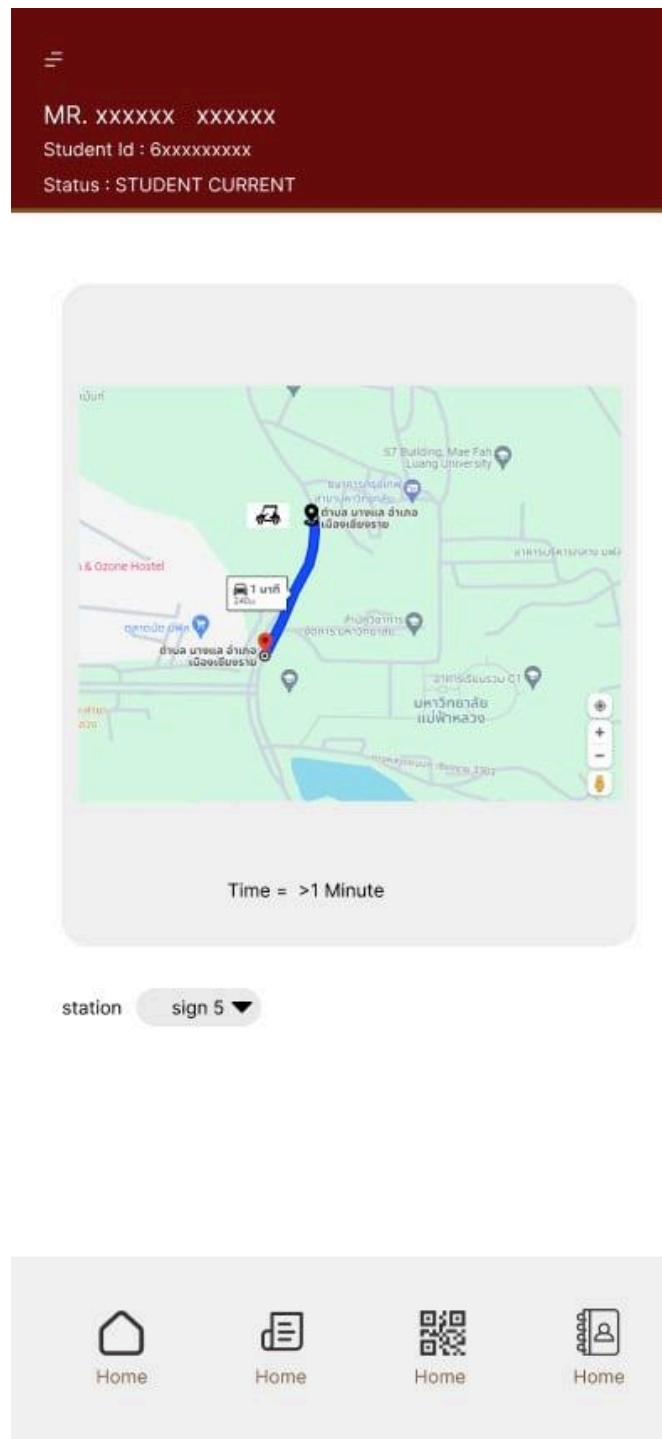


**Figure 3.7 Select a station**

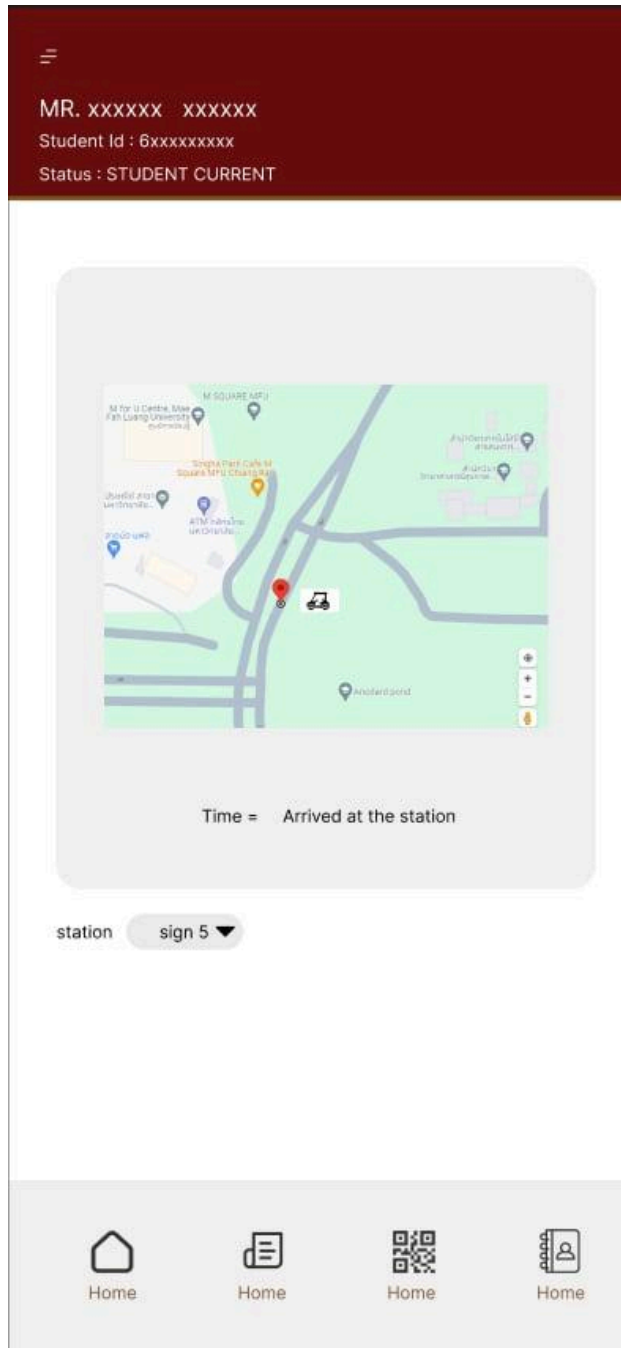


**Figure 3.8 Show distance and time**





**Figure 3.9 Show distance and time  
and App notifications**



**Figure 3.10 Arrive at the station**

## **Summary of Project**

# **Summary of Project:**

## **GPS System for jem car**

**Our project aims to create a GPS system for Jem car, intended to be used by students who need to use but cannot predict the exact timing. We have encountered challenges in achieving our goals. One issue is the difficulty in uploading latitude and longitude data to the website. Additionally, there is a problem with James's vehicles lacking Wi-Fi connectivity.**

**However, we have implemented a notification Line system that alerts users when the vehicle approaches the designated points. Despite these challenges, we have achieved the expected results as planned**

## **REFERENCES**

## REFERENCES

[1] Theory of Relativity with GPS Systems

<https://www.narit.or.th/index.php/astronomy-article/222-relating-gps-systems?fbclid>

[2] The distance between the satellite and the receiver can be measured using

<https://www.global5thailand.com/thai/gps.htm>

[3] Related Work

<file:///C:/Users/Windows%2010%20Pro/Downloads/mako,+Journal+manager,+85-102.pdf>

# **APPENDIX**

# APPENDIX

## SOURCE CODE

esp32

```
#define RXD2 18
#define TXD2 19
#include <TridentTD_LineNotify.h>

#define SSID      "FK"    // บรรทัดที่ 11 ให้ใส่ ชื่อ Wifi ที่จะเชื่อมต่อ
#define PASSWORD  "0660198989" // บรรทัดที่ 12 ใส่ รหัส Wifi
#define LINE_TOKEN "m442TmiRt0XHUKsowym12MuwtU7OyEqE2s9zkwUv220"
// บรรทัดที่ 13 ใส่ รหัส TOKEN ที่ได้มาจากช่างบน
double targetLatitude = 20.049770;
void setup() {
    // initialize digital pin LED_BUILTIN as an output.
    Serial.begin(115200);
    Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2);
    WiFi.begin(SSID, PASSWORD);
    Serial.printf("WiFi connecting to %s\n", SSID);
    while (WiFi.status() != WL_CONNECTED) {
        Serial.print(".");
        delay(400);
    }
    Serial.printf("\nWiFi connected\nIP : ");
    Serial.println(WiFi.localIP());
    // กำหนด Line Token
    LINE.setToken(LINE_TOKEN);
    // ตัวอย่างส่งข้อความ
    LINE.notify ("สวัสดี!!");
}

void loop() {
    if (Serial2.readString()){
        String receivedData = Serial2.readStringUntil('\n');
        double newLatitude = receivedData.toDouble();
        //targetLatitude = Serial2.readString();
        if (newLatitude != targetLatitude) {
            targetLatitude = newLatitude;
            LINE.notify("ถึงป้ายรอรถแล้ว!!!");
        }
    }
    return
    delay(100);
}
```



## GPS module 6M

```
#include <TinyGPS++.h>
#include <SoftwareSerial.h>

static const int RXPin = 4, TXPin = 3;
static const uint32_t GPSBaud = 9600;

// The TinyGPS++ object
TinyGPSPlus gps;

// The serial connection to the GPS device
SoftwareSerial ss(RXPin, TXPin);

void setup() {
  Serial.begin(9600);
  ss.begin(GPSBaud);
}

void loop() {
  // This sketch displays information every time a new sentence is
  // correctly encoded.
  while (ss.available() > 0) {
    gps.encode(ss.read());
    if (gps.location.isUpdated()) {
      // Latitude in degrees (double)
      Serial.print(gps.location.lat(), 6);
      Serial.print(", ");
      Serial.println(gps.location.lng(), 6);
    }
  }
}
```